

Two-Photon Physics at the $LH\mu C$

High-energy $\gamma\gamma$ interactions in μp collisions

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(on behalf of HELP4CERN Study Group)

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- 1 LH μ C and the $\gamma\gamma$ opportunity
- 2 Equivalent Photon Approximation and $\gamma\gamma$ luminosity
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- 5 Heavy and BSM final states
- 6 Experimental opportunities and outlook

Why a muon-hadron collider for two-photon physics?

- A muon-hadron collider combines a **clean lepton-hadron initial state** with a multi-TeV centre-of-mass energy.
- Replacing the electron beam by a muon beam strongly relaxes the synchrotron-radiation limitation, making much higher lepton-beam energies possible.
- The μp environment remains experimentally cleaner than pp for many exclusive and semi-exclusive final states: negligible pile-up is a central advantage of the lepton-hadron concept.
- Both beams radiate photons. Their electromagnetic fields therefore create a high-energy $\gamma\gamma$ collider *inside* the μp machine.

Take-home message

LH μ C combines a high-energy photon source from the muon beam with the electromagnetic photon field of the proton, opening a distinctive high-energy $\gamma\gamma$ programme.

Physics motivation adapted from the HELP4CERN / DIS2026 decks and the LHeC $\gamma\gamma$ study.

LH μ C@CERN: benchmark working points



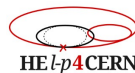
- **Workshop focus:** $E_\mu = 500$ GeV and $E_p = 7$ TeV,

$$\sqrt{s_{\mu p}} \simeq 3.7 \text{ TeV}, \quad \mathcal{L}_{\text{int}} \simeq 250 \text{ fb}^{-1}.$$

- Higher-energy HELP4CERN working points remain useful for scaling studies:
 - $E_\mu = 1 \text{ TeV} \Rightarrow \sqrt{s} \simeq 5.3 \text{ TeV}$,
 - $E_\mu = 2 \text{ TeV} \Rightarrow \sqrt{s} \simeq 7.5 \text{ TeV}$.
- Already the 500 GeV option enters a qualitatively new $\gamma\gamma$ mass regime compared with LHeC.

Working points

	E_ℓ	$\sqrt{s}$
LHeC	50 GeV	1.2 TeV
LH μ C	500 GeV	3.7 TeV
LH μ C	1 TeV	5.3 TeV
LH μ C	2 TeV	7.5 TeV

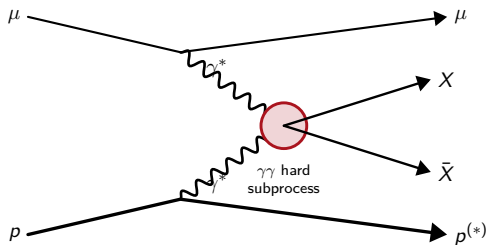


This talk

The quantitative $\gamma\gamma$ benchmarks shown below emphasise the 500 GeV μ beam whenever dedicated results exist.

HELP4CERN benchmark set used in the DIS2026 HET study.

A high-energy $\gamma\gamma$ collider inside μp collisions



Both beams act as photon sources; the central system
is produced by $\gamma\gamma$ fusion.

Factorised picture

$$\mu \rightarrow \mu + \gamma^*, \quad p \rightarrow p^{(*)} + \gamma^*,$$

$$\gamma\gamma \rightarrow X, \quad \mu p \rightarrow \mu + X + p^{(*)}.$$

- **Elastic:** proton remains intact.
- **Tagged elastic:** intact proton measured near the beam.
- **Dissociative:** proton breaks into a system of mass M_N .

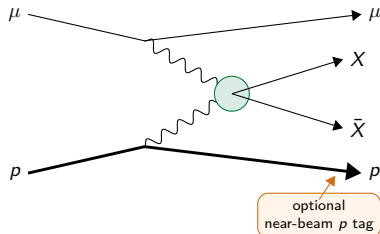
Key point

One parent μp data set contains several distinct $\gamma\gamma$ topologies with different photon virtualities and experimental handles.

Two-photon production topologies: elastic, tagged and proton-dissociative

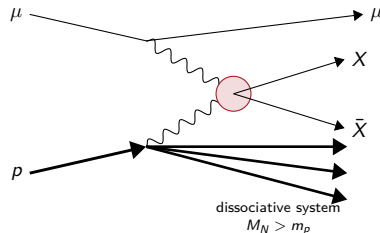


Elastic / tagged elastic



- Proton electromagnetic form factors control the elastic flux.
- A near-beam tag can over-constrain event kinematics.

Proton-dissociative



- Inelastic flux is described through proton structure functions.
- Larger M_N and Q_p^2 increase the accessible luminosity but also modelling dependence.

Topology definitions follow the LHeC $\gamma\gamma$ study; the same classification carries directly to μp .

Equivalent Photon Approximation: factorising the μp cross section

EPA master relation

$$\sigma_{\mu p} = \int dy_\mu dy_p \Phi_\mu(y_\mu) \Phi_p(y_p) \sigma_{\gamma\gamma}(W) = \int dW S_{\gamma\gamma}(W) \sigma_{\gamma\gamma}(W)$$

- Φ_μ and Φ_p are the equivalent-photon fluxes radiated by the two beams.
- The hard subprocess $\gamma\gamma \rightarrow X$ factorises from photon emission, in close analogy with partonic convolutions.
- $S_{\gamma\gamma}(W)$ packages the two fluxes into the effective two-photon luminosity spectrum.

EPA formalism: yy_lhep / draft_yy_lhep.

Kinematic variables

$$y_\mu = \frac{E_\gamma^{(\mu)}}{E_\mu}, \quad y_p = \frac{E_\gamma^{(p)}}{E_p},$$

$$W \equiv W_{\gamma\gamma}.$$

Equivalent photon fluxes from the muon and proton

Generic EPA flux

$$\Phi_\gamma(y) = \frac{\alpha}{\pi y} \int \frac{dQ^2}{Q^2} \left[(1-y) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y^2}{2} F_M(Q^2) \right]$$

$$Q_{\min}^2 = \frac{M^2 y^2}{1-y}.$$

- Point-like lepton: $F_E = F_M = 1$ and $M = m_\mu$.
- The muon mass raises $Q_{\min}^2$ relative to an electron at the same photon energy fraction y .

Elastic proton emission

$$F_M = G_M^2,$$

$$F_E = \frac{4m_p^2 G_E^2 + Q^2 G_M^2}{4m_p^2 + Q^2}.$$

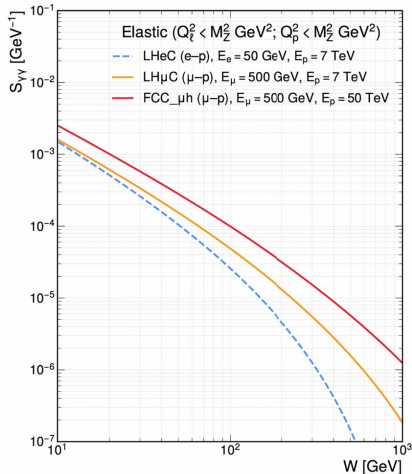
The dipole form factors suppress large Q_p^2 .

Proton dissociation

For $p \rightarrow X$, structure functions replace elastic form factors and the flux depends on the dissociative mass M_N .

Photon-flux formula and proton form factors: theory input of yy_lhec / draft_yy_lhec.

The $\gamma\gamma$ luminosity spectrum $S_{\gamma\gamma}(W)$



$S_{\gamma\gamma}$ tells us

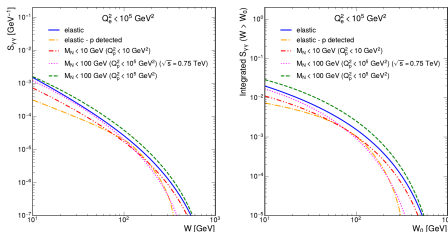
The fraction of the parent lepton-proton luminosity available for two-photon collisions at invariant mass W .

- The 500 GeV LH μ C spectrum extends well beyond the LHeC reach.
- A harder spectrum benefits the WW , ZZ , $t\bar{t}$ and charged-BSM thresholds.
- Energy-growing EFT effects are naturally amplified in the high- W tail.

LH μ C message: the gain is not only in total statistics; the $\gamma\gamma$ spectrum is substantially harder.

Existing HELP4CERN elastic-luminosity comparison; orange: $E_\mu = 500$ GeV, $E_p = 7$ TeV.

Elastic vs. inelastic $\gamma\gamma$ luminosity and forward proton tagging



LHeC reference study

The detailed calculation separates elastic, tagged elastic ($0.01 < y_p < 0.1$), and proton-dissociative photon emission for several M_N and Q_p^2 ranges.

- Larger allowed dissociative masses increase the usable $\gamma\gamma$ luminosity.
- A forward proton tag improves kinematic closure, especially for final states with neutrinos.
- Low pile-up means proton tagging need not be mandatory for every clean channel.

LHeC $\gamma\gamma$ paper: luminosity spectrum and near-beam tagging discussion; topology logic transfers to LH μ C.

Beyond the simplest EPA: photon virtuality and exact $W_{\gamma\gamma}$ kinematics

High-energy approximation

$$W^2 \simeq y_\ell y_p s.$$

Exact elastic relation in the updated study

$$W^2 = -Q_\ell^2 + y_\ell y_p s$$

The paper writes $\ell = e$; the same lepton-side kinematics applies after $e \rightarrow \mu$.

Updated exact-kinematics study: draft_yy_lhec, Appendices C–E.

- The approximation is excellent when photon virtualities are negligible relative to W^2 .
- For proton dissociation, the exact expression also depends on Q_p^2 and M_N .
- Elastic EPA agrees at percent level with matrix-element generators once the proper W definition is used; inelastic modelling is more sensitive to the dissociative phase space.

Precision lesson

At sizeable Q^2 , correct $W_{\gamma\gamma}$ reconstruction is part of the physics model, not a cosmetic refinement.

From photon luminosity to physics: $\gamma\gamma \rightarrow X$



Collider rate

$$\sigma(\mu p \rightarrow \mu X p^{(*)}) = S_{\gamma\gamma} \otimes \sigma(\gamma\gamma \rightarrow X).$$

$\ell^+\ell^-$
calibration

$\tau^+\tau^-$
dipole
moments

W^+W^-
gauge
structure

ZZ
loop / nQGC

$t\bar{t}$
heavy
threshold

charged BSM
pairs

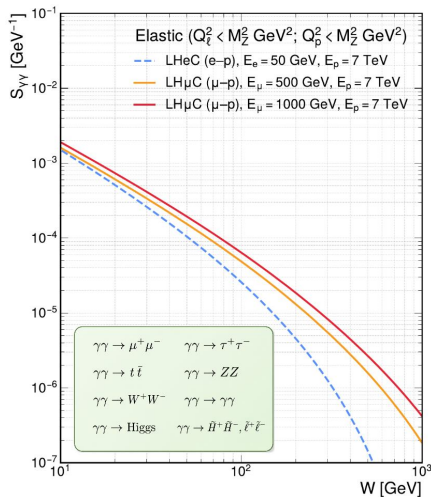
$\gamma\gamma$
light-by-light

H
loop-induced

Take-home message

The same photon luminosity supports precision QED, electroweak and loop physics, heavy thresholds, and direct searches for charged new states.

The high-energy $\gamma\gamma$ physics landscape at LH μ C



Existing HELP4CERN threshold summary, adapted to the dedicated two-photon narrative.

- $ll, \tau\tau$: high-rate QED calibration and dipole probes.
- WW : sizeable SM rate and direct $\gamma\gamma WW$ coupling.
- ZZ : loop-induced SM benchmark and neutral quartic couplings.
- $t\bar{t}$: heavy threshold and top electromagnetic structure.
- Charged BSM pairs: production controlled mainly by electric charge and mass.
- $\gamma\gamma$ and H : loop-induced quantum processes and complementary searches.

$\gamma\gamma \rightarrow \ell^+ \ell^-$: a standard candle for two-photon physics



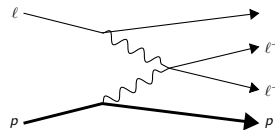
- e^+e^- and $\mu^+\mu^-$ provide clean control samples for the complete $\gamma\gamma$ programme.
- They test EPA modelling, acoplanarity and elastic/inelastic separation.
- At LHeC, elastic EPA predictions agree with GRAPE and CepGen/LPAIR at the percent level for matched cuts and exact W kinematics.

Elastic $\mu^+\mu^-$ benchmark at LHeC

$$Q_{e,p}^2 < 10 \text{ GeV}^2$$

	$W_0 = 10 \text{ GeV}$	$W_0 = 100 \text{ GeV}$
EPA	120.12 pb	0.193 pb
CepGen/LPAIR	118.31 pb	0.191 pb
GRAPE (BH)	121.37 pb	0.193 pb

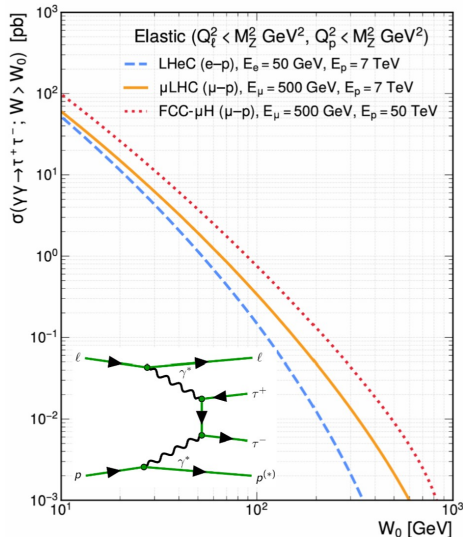
LHeC $\gamma\gamma$ paper, Table 1 and calibration-candle discussion.



Why it matters for LH μ C

The same QED process becomes a high-statistics calibration candle for the harder μp photon spectrum and any forward-tagging strategy.

$\gamma\gamma \rightarrow \tau^+\tau^-$: a high-statistics tau factory



Elastic cross sections from the existing HELP4CERN two-photon calculations.

Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 50.37 \text{ pb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 58.56 \text{ pb}$$

- Large statistics make $\tau\tau$ a natural precision channel.
- LH μ C extends the accessible $M_{\tau\tau}$ tail.
- The hard tail enhances sensitivity to non-standard $\gamma\tau\tau$ interactions.

Anomalous τ electromagnetic moments: the $\gamma\tau\tau$ vertex

General electromagnetic vertex

$$\Gamma^\mu(q^2) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m_\tau} F_2(q^2) + \sigma^{\mu\nu}q_\nu \gamma_5 F_3(q^2),$$

$$F_2(0) = a_\tau, \quad F_3(0) = -\frac{2m_\tau}{e} d_\tau.$$

- a_τ : CP-even anomalous magnetic dipole moment.
- d_τ : CP-odd electric dipole moment.
- $\gamma\gamma \rightarrow \tau^+\tau^-$ probes the vertex directly.

Dedicated anomalous- τ study, April 2026; updated benchmark on the next slide.

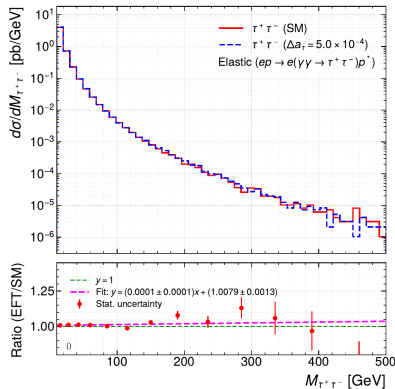
SMEFT interpretation

Gauge-invariant tau dipoles:

$$\bar{L}_\tau \sigma^{\mu\nu} \tau_R H \left(\frac{C_{\tau B}}{\Lambda^2} B_{\mu\nu} + \frac{C_{\tau W}}{\Lambda^2} W_{\mu\nu} \right) + \text{h.c.}$$

Real and imaginary coefficient combinations map respectively to magnetic and electric dipole deformations after electroweak symmetry breaking.

Probing τ dipole moments in the high- $M_{\tau\tau}$ tail



Updated draft_yy_lhec: $\Delta a_\tau = 5 \times 10^{-4}$ and SMEFT translation.

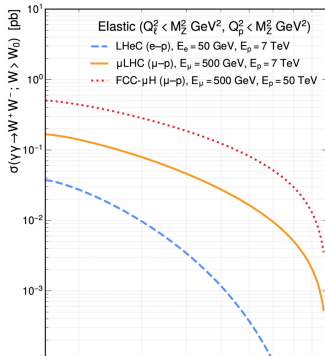
Updated paper benchmark

The latest draft replaces the previous illustrative point by

$$\Delta a_\tau = 5.0 \times 10^{-4}.$$

- Shown: updated **LHeC** elastic reference calculation from the paper draft.
- The EFT/SM ratio exposes the growing shape distortion in the hard $M_{\tau\tau}$ region.
- The same SMEFT fit should be repeated for $\text{LH}\mu\text{C}$, whose harder spectrum gives more leverage at high mass.

$\gamma\gamma \rightarrow W^+W^-$: a clean electroweak laboratory



Existing HELP4CERN elastic WW calculation; orange curve: $\text{LH}\mu\text{C}@3.7$ TeV.

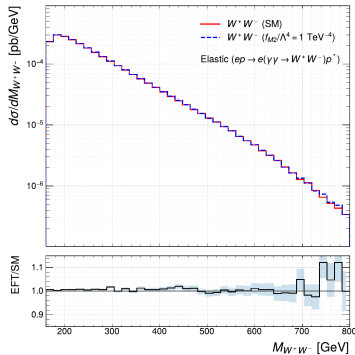
Elastic rates

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.037 \text{ pb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.166 \text{ pb}$$

- The W threshold lies inside the high-luminosity $\gamma\gamma$ regime.
- The process directly tests the SM $\gamma\gamma WW$ gauge structure.
- The high- M_{WW} tail is a clean lever arm for dimension-8 quartic-gauge operators.

Anomalous quartic gauge couplings in $\gamma\gamma \rightarrow W^+W^-$



Dimension-8 EFT

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i}{\Lambda^4} \mathcal{O}_i,$$

$$\frac{f_{M2}}{\Lambda^4} = 1 \text{ TeV}^{-4}.$$

- Shown: updated **LHeC** elastic reference from the draft.
- EFT effects are tiny at low M_{WW} and grow in the hard tail.
- $\text{LH}\mu\text{C}$ supplies the harder $W_{\gamma\gamma}$ spectrum needed to extend this strategy quantitatively.

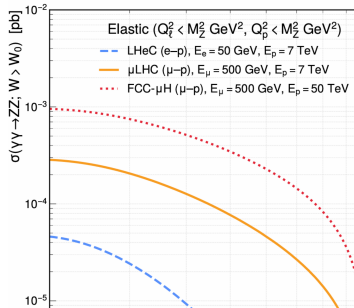
Scope: no detector-level $\text{LH}\mu\text{C}$ exclusion is

claimed here; this is a defined next **HELP4CERN**

study.

Updated draft_yy_lhec: WW aQGC benchmark and EFT/SM ratio.

$\gamma\gamma \rightarrow ZZ$: a loop-induced Standard-Model channel



- In the SM, $\gamma\gamma \rightarrow ZZ$ is absent at tree level and proceeds through loops.
- The small SM rate makes the channel a clean window on anomalous neutral quartic interactions.
- LH μ C extends the accessible high- M_{ZZ} regime.

Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.046 \text{ fb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.284 \text{ fb}$$

Existing HELP4CERN elastic ZZ calculation; updated draft motivates the neutral-QGC study.

Neutral quartic gauge couplings in $\gamma\gamma \rightarrow ZZ$



Rate model used in the updated study

$$\sigma_{\text{tot}} \simeq \sigma_{\text{SM}}^{\text{loop}} + \sigma_{\text{AQGC}}^{\text{tree}}.$$

LHeC tagged-elastic reference

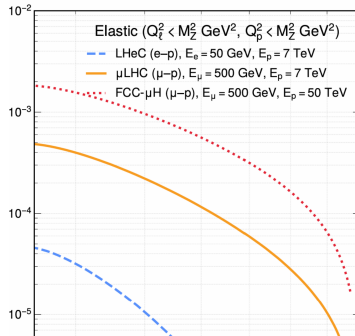
$\sigma_{\text{SM}} = 3.340 \times 10^{-5}$ pb; raw events assume 1 ab^{-1} before Z decays and detector effects.

f_{M2}/Λ^4	σ_{AQGC} [pb]	events	enhancement
0.1	1.17×10^{-8}	0.012	1.0004
1.0	1.17×10^{-6}	1.17	1.035
5.0	2.93×10^{-5}	29.3	1.88

Updated draft_yy_lhec, neutral aQGC section.

- SM is loop induced; the quoted AQGC contribution is tree level.
- The table shows how an energy-growing neutral interaction can become competitive with a rare SM channel.
- These are **LHeC reference numbers**, not $\text{LH}\mu\text{C}$ projected limits; the harder $\text{LH}\mu\text{C}$ spectrum motivates the direct extension.

Heavy final states: $\gamma\gamma \rightarrow t\bar{t}$



Existing HELP4CERN elastic $\gamma\gamma \rightarrow t\bar{t}$ calculation.

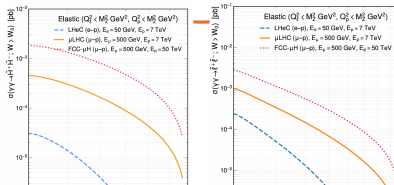
Elastic benchmark

$$\sigma_{\text{el}}^{\text{LHeC}} \simeq 0.046 \text{ fb}$$

$$\sigma_{\text{el}}^{\text{LH}\mu\text{C}(3.7)} \simeq 0.486 \text{ fb}$$

- This is exclusive $\gamma\gamma \rightarrow t\bar{t}$ — **not** the much larger $\gamma g \rightarrow t\bar{t}$ photoproduction channel.
- The $2m_t$ threshold probes the hard end of the photon spectrum.
- The channel accesses the top electromagnetic interaction and high-energy new physics.

Beyond the Standard Model: charged-particle pair production in $\gamma\gamma$ collisions



Illustrative

100 GeV scalar (slepton) and fermion (higgsino) reference curves.

- Representative channels:

$$\gamma\gamma \rightarrow \tilde{\ell}^+ \tilde{\ell}^-, \quad \gamma\gamma \rightarrow \tilde{\chi}^+ \tilde{\chi}^-.$$

- Rates are governed mainly by electric charge and mass, complementary to strongly interacting searches.
- Forward kinematic closure is attractive for compressed spectra with soft visible products and missing momentum.

Updated interpretation

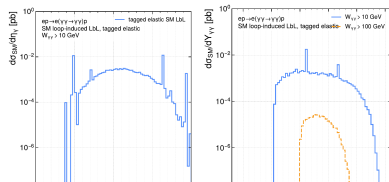
The draft notes that 100 GeV is now illustrative. Current constraints motivate roughly 300 GeV sleptons and 250–300 GeV higgsino benchmarks; LHeC rates near those thresholds are tiny, making a dedicated LH μ C re-evaluation especially relevant.

yy_lhec SUSY section + updated compressed-SUSY discussion in draft_yy_lhec.

Light-by-light scattering, Higgs production, and other $\gamma\gamma$ opportunities

Mature calculation: $\gamma\gamma \rightarrow \gamma\gamma$

SM light-by-light scattering is loop induced; all charged SM particles contribute to the box amplitude.



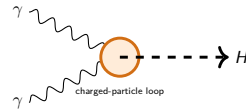
Updated

LHeC tagged-elastic SM pseudorapidity distributions.

Updated draft_yy_lhec LbL section; Higgs-via- $\gamma\gamma$ section in yy_lhec / draft_yy_lhec.

Exploratory: $\gamma\gamma \rightarrow H$

The Higgs channel is loop induced and rare, but it probes the effective $H\gamma\gamma$ interaction in an exceptionally clean initial state.



LH μ C opportunity

The harder photon spectrum extends the high-mass LbL region and other loop-induced or resonant searches.

What does two-photon physics require from the $\text{LH}\mu\text{C}$ experiment?



- **Wide central tracking and lepton acceptance**, including robust τ reconstruction.
- **Efficient $W/Z/t$ reconstruction** for heavy $\gamma\gamma$ final states.
- **Exclusivity and acoplanarity** observables with sensitivity to low extra activity.
- **Missing-momentum performance** for neutrinos and compressed BSM signatures.
- **Scattered-muon measurement** to constrain the lepton-side photon kinematics.
- **Forward proton tagging where feasible**, together with control of proton dissociation.
- Exploit the defining advantage: the **clean, low-pile-up μp environment**.

Figure placeholder

Suggested figure: $\text{LH}\mu\text{C}$ interaction-region schematic with central detector, scattered-muon acceptance and a near-beam proton tag.

Physics–detector connection

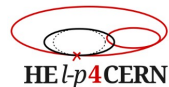
The $\gamma\gamma$ programme links central reconstruction, lepton tagging and near-beam instrumentation directly to the usable photon luminosity and kinematic closure.

Summary and outlook



- $\text{LH}\mu\text{C}$ turns μp collisions into a **high-energy $\gamma\gamma$ laboratory** with a substantially harder photon spectrum than the LHeC benchmark.
- EPA and $S_{\gamma\gamma}$ connect parent μp luminosity transparently to elastic and proton-dissociative two-photon subprocesses.
- $\tau^+\tau^-$ and W^+W^- provide high-statistics precision/EFT channels; sensitivity lives in their hard invariant-mass tails.
- Loop-induced ZZ , exclusive $t\bar{t}$, light-by-light scattering, H , and charged BSM pairs extend the programme to rare quantum processes and heavy thresholds.

Bottom line: high-energy $\gamma\gamma$ physics is one of the distinctive physics opportunities of $\text{LH}\mu\text{C}$.



Thank you!

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